

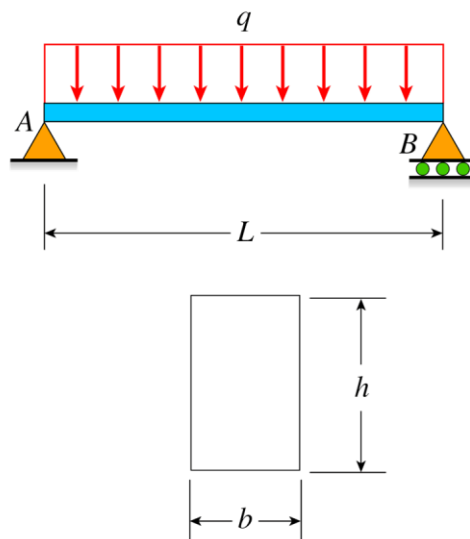
Submission: Please submit the scanned/typed document to Canvas.

1. A 1-meter-long, simply supported copper beam ($E=117$ GPa) carries uniformly distributed load q . The maximum deflection is measured as 1.5 mm.

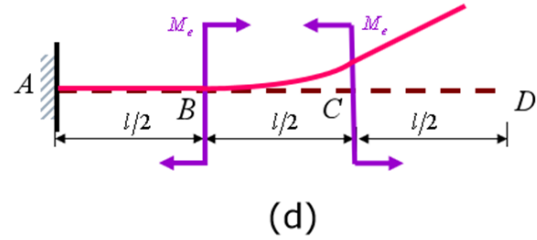
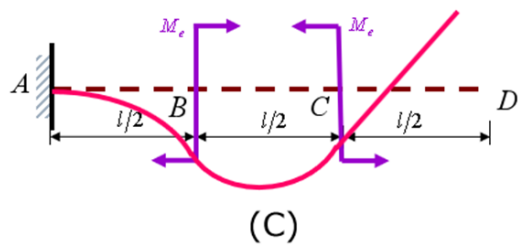
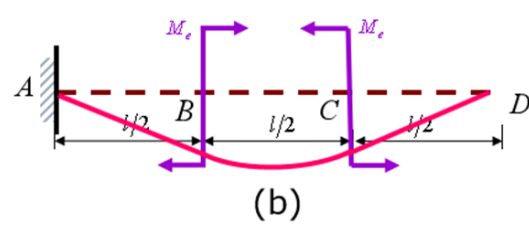
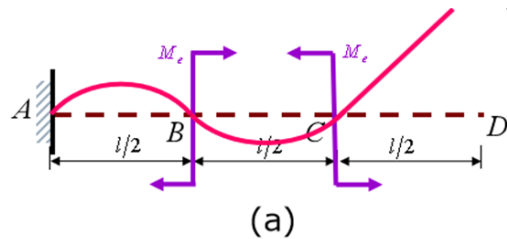
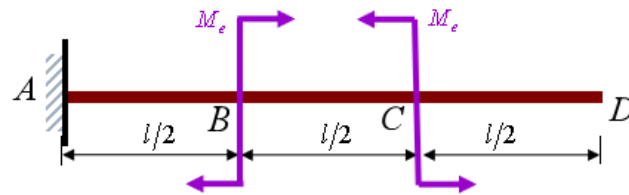
(a) Calculate the magnitude of the distributed load q if the beam has a rectangular cross section (width $b=20$ mm, height $h=40$ mm).

(b) If instead the beam has circular cross section and $q=500$ N/m, calculate the radius r of the cross section.

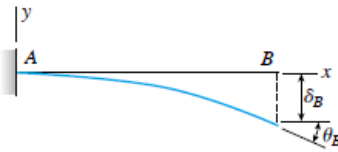
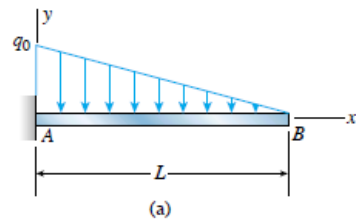
Neglect the weight of the beam.



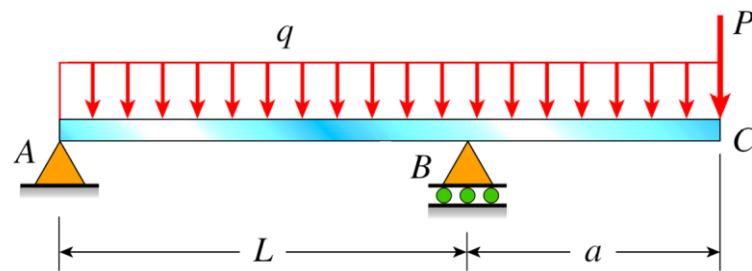
2. The cantilever beam is subjected to the loading shown in the figure. Regarding the beam's deflection curve, four possible answers are given. Please analyze and determine which one is correct.



3. A cantilever beam AB supporting a triangularly distributed load of maximum intensity q_0 is shown in the figure. Derive the equation of the deflection curve and then obtain formulas for the deflection δ_B and deflection angle θ_B at the free end. Use the second-order differential equation of the deflection curve.



4. Beam ABC is loaded by a uniform load q and point load P at joint C. Using the method of superposition, calculate the deflection at joint C. Assume that $L = 4$ m, $a = 2$ m, $q = 15$ kN/m, $P = 7.5$ kN, $E = 200$ GPa, and $I = 70.8 \times 10^6$ mm⁴.



5. Beams AB and BC are pin-connected at B, with ends A and C fixed. Both beams have the same flexural rigidity EI . The applied concentrated force is $F=40$ kN, and the uniformly distributed load is $q=20$ kN/m. (a) calculate the vertical reaction forces and reaction bending moments on A and C ends. (b) Draw the SFD and BMD for the beam. (Hint: There are no horizontal reactions (loading is vertical only). The structure is statically indeterminate.)

